

# Explosive Detection within and Across Baggage



## 01 Introduction and Objectives

RayGuard addresses two complementary screening problems: concealed threats inside a single bag and component evidence distributed across related packages. Clutter, overlap and small targets complicate both. Guided by the SDP proposal, we combine localization, structured evidence and operator review, drawing on IEDXray and FALCON [1–3].

### P1 | Concealed threats within one bag

Identify electronic devices and localize suspicious regions despite concealment and overlap. Our objective is to connect each region to its host and support an interpretable review. IEDXray provides 17,360 RGB scans and separate device, generic-explosive and specific-explosive detection tasks [1,3].

## 02 Methods and working demonstration

### P1 - SINGLE-BAGGAGE LOCALIZATION

#### 1 Image intake

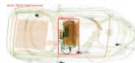


#### 2 YOLOv10-M

Selected task checkpoint

Device Generic Specific

#### 3 Localized evidence



### CURRENT DETECTOR BASELINE

#### 4 Operator review

Box • label • score  
Confidence: 0.970  
Overlay + JSON export

Excerpt: independent task detectors with original-image coordinates. Next: evaluate device-conditioned localization and explicit host–region association. The saved demonstration is a training image, not an accuracy estimate.

### P2 - MULTI-PACKAGE REASONING



#### Component grounding

Regions + masks + features  
Retain scan and region IDs

#### Score-based late fusion

Multi-input model

### PROPOSED EXTENSION

#### Joint case review

Traceable case evidence  
Human assessment

Verified case membership

Compare fusion strategies on held-out cases, retaining related scans together. Assess case recall, false alarms and missing-scan sensitivity; group-level results remain to be measured [2,3].

## 03 Single-bag evidence

### (a) Laptop concealment - Fig. 6(a)



### (b) Concealed target - Fig. 6(f)



Ground truth Faster R-CNN YOLOv10  
Cascade R-CNN DETR  
Same-scan detail crops; original published boxes [1].

AP@[0.50:0.95] (%) - Tables 2–4 [1]

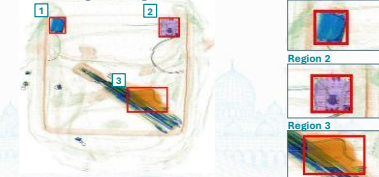
| Model          | Device | Generic | Specific |
|----------------|--------|---------|----------|
| Grounding DINO | 69.0   | 48.9    | 43.5     |
| YOLOv10-M      | 67.0   | 35.1    | 36.8     |
| RTMDet         | 61.3   | 28.9    | 29.0     |

Device recognition is easier than concealed-threat localization. A device box alone does not establish a threat.

## 04 Component Reasoning

### Query: ground functionally related components

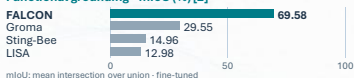
One scan - three grounded regions



Published box Colored mask

FALCON grounds three candidate regions within one image [2, Fig. 9]. This motivates cross-package reasoning; multi-bag performance remains to be evaluated.

### Functional grounding - mIoU (%) [2]



[1] Takle et al. Scientific Data 13, 962 (2026). doi:10.1038/s41597-026-07263-7.

[2] Michael et al. FALCON. arXiv:2508.25701v2 [2026], Table 2; Fig. 9, 915, Figures [1,2]. CC BY 4.0.

[3] Werghi, AI-Powered Explosive Detection in X-ray Scans, SDP proposal and advisor meeting records (2026).